

CS 203: Software Tools & Techniques for AI
IIT Gandhinagar
Sem-II - 2024-25

LAB 11

Lead TA: Himanshu Beniwal

Total marks: 100

☐ **Submission deadline: Sunday, 20/04/2025 11:59 PM**

Submission guidelines:

1. Code should be added to a GitHub repository, and the **repository details should be shared in the pdf.**
2. **Submit the PDF showing screenshots** of all steps involved in the following code.

Note: Submitting this assignment solution confirms that you will follow the IITGN's honor code. We shall strictly penalize the submissions containing plagiarized text/code.

Objective:

This assignment aims to learn about model quantization.

1. Dataset Preparation (10%)

- Load the [training dataset](#) and [test data](#).
- Use 20% of the training dataset as the validation set.

2. Construct a Multi-Layer Perceptron (MLP) model. (10%)

- The parameter should be with:
 - hidden_sizes=[512, 256, 128, 64]
 - Output should have two labels.
 - With the following architecture:

```

Model Summary:
=====
Input Size: 10000
Layer 1: Linear(10000, 512)
  Parameters: 5,120,512
  Activation: ReLU + Dropout(0.3)
Layer 2: Linear(512, 256)
  Parameters: 131,328
  Activation: ReLU + Dropout(0.3)
Layer 3: Linear(256, 128)
  Parameters: 32,896
  Activation: ReLU + Dropout(0.3)
Layer 4: Linear(128, 64)
  Parameters: 8,256
  Activation: ReLU + Dropout(0.3)
Layer 5: Linear(64, 2)

```

- Count the number of trainable parameters in the model using the automated function ([reference](#)).

3. Train the model with 10 epochs and create the best-performing model (checkpoint.pt) (10%)

- Plot the validation accuracy + loss (epochs vs accuracy-loss).

4. Now use:

1. Dynamic Quantization with INT4 (Link: [here](#)) (20%)

- Use the `torch.quantization.quantize_dynamic()`

2. Half precision (20%)

- Use the `.half()` function. (Reference: [here](#))

5. Fill the table for different quantization techniques. (30%)

S.I.	Model Name	Accuracy (Out of 100)	Storage (In MB)	Inference time (In ms)
1.	Original			
2.	Dynamic			
3.	Half			